

ME1103

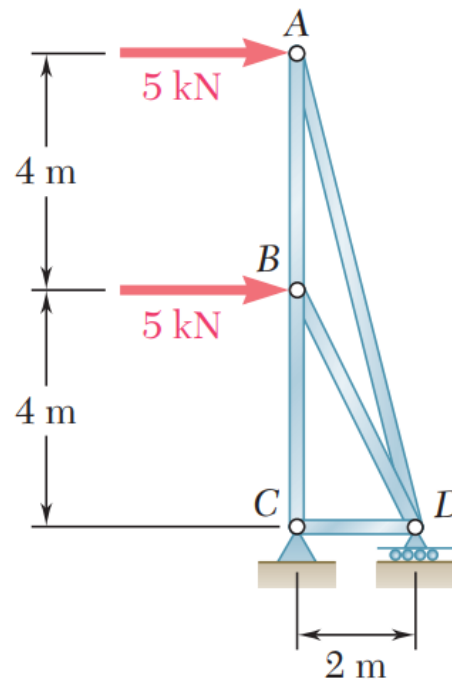
Principles of Mechanics and Materials

PROBLEM SET 3

Question 1

Using the method of joints, determine the force in each member of the truss shown. State whether each member is in tension or compression.

Answer: $F_{AB} = 20 \text{ kN}$; $F_{AD} = -20.6 \text{ kN}$;
 $F_{BD} = -11.2 \text{ kN}$; $F_{BC} = 30 \text{ kN}$; $F_{CD} = 10 \text{ kN}$.



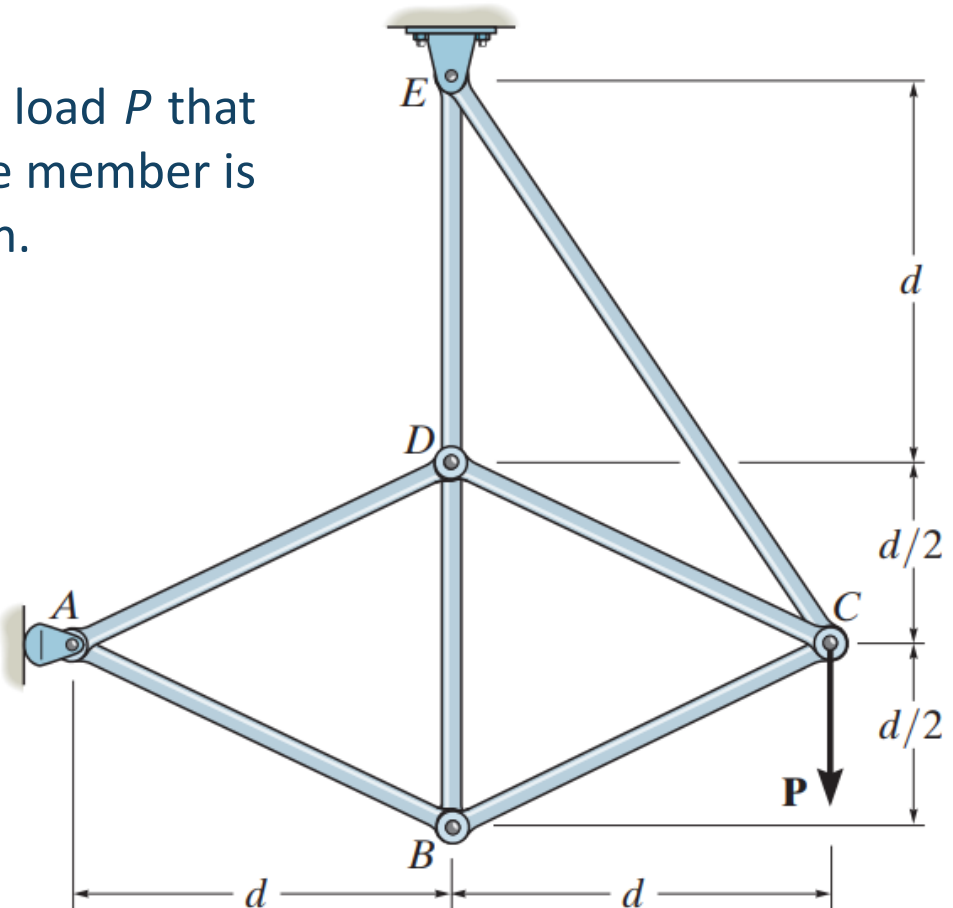
Question 2

Given that the maximum allowable tensile and compressive force in the members of the truss are 5 kN and 3 kN, respectively.

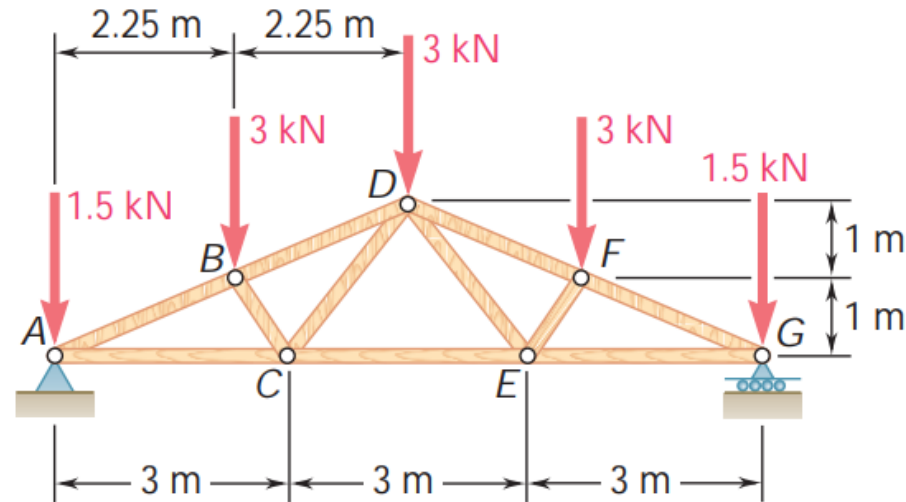
Determine the maximum magnitude of the load P that can be applied to the truss, and which of the member is the weakest link under this loading condition.

Answer: $P_{max} = 4.16 \text{ kN}$

and member EC is the weakest link.



Question 3



Using the method of joints, determine the force in each member of the Fink roof truss shown. And also using the method of section, find the forces in members CD and CE .

Answer: $F_{AB} = -11.08 \text{ kN}$; $F_{AC} = 10.13 \text{ kN}$;

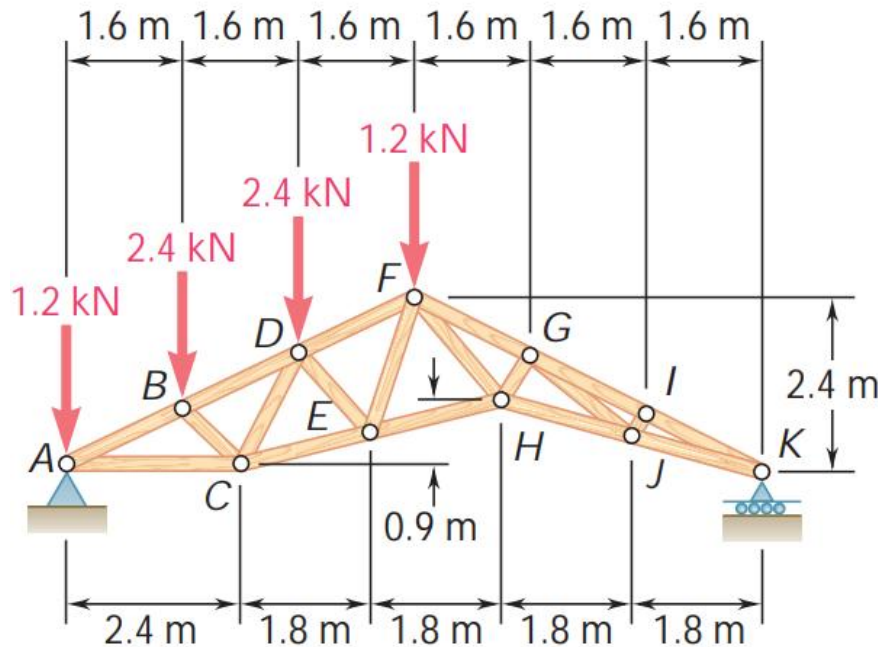
$F_{BC} = -2.81 \text{ kN}$; $F_{BD} = -9.23 \text{ kN}$;

$F_{CD} = 2.81 \text{ kN}$; $F_{CE} = 6.75 \text{ kN}$.

Question 4

For the vaulted roof truss shown, determine:

- Zero-force members in this assembly,
- Using method of joints, the force in members connecting joint C, and
- Using method of section, the force in member EH .



Answer:

(a) Zero-force members are IJ , GJ and GH .

(b) $F_{AC} = 8.40 \text{ kN (T)}$; $F_{BC} = 2.26 \text{ kN (C)}$;

$F_{CD} = 0.128 \text{ kN (C)}$; $F_{CE} = 7.07 \text{ kN (T)}$.

(c) $F_{EH} = 4.948 \text{ kN (T)}$